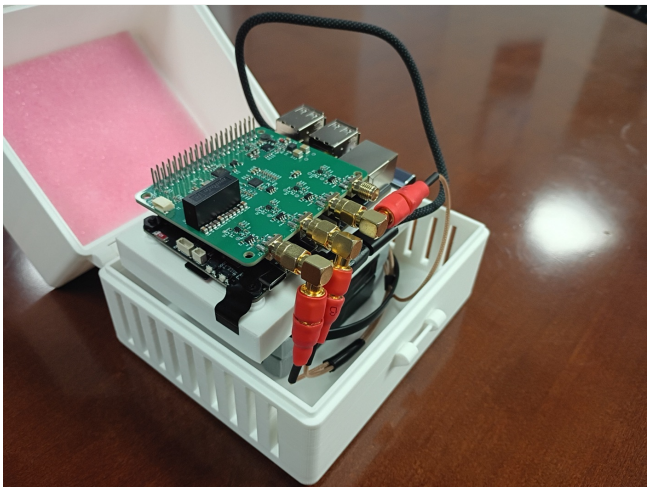
gLOWCOST detector quick guide

ESP32-P4 field start-up and Wi-Fi setup

Start a run

1. Insert a FAT32-formatted microSD card before power-up.
2. Connect the detector to the USB power bank. It starts automatically.
3. Leave the detector still while the FPGA, DAC, and high voltage are configured.
4. Wait through the 10 second HV settling period. Counting and SD logging then begin.



Detector opened for setup

Open the detector page

Turn mobile data off so the phone stays connected to the detector network.

Wi-Fi	Password	Web page
MuonReadout	**glowcost**	**192.168.4.1**

Before the measurement

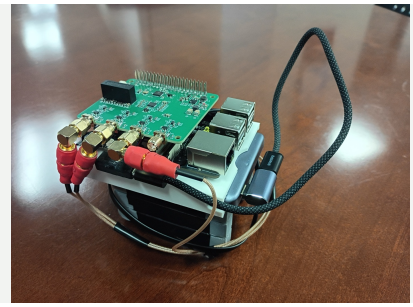
- [ ] SD Card shows Ready.
- [ ] Browser time has synced to the detector.
- [ ] One-minute coincidence counts look reasonable.
- [ ] Flight number, location, or run label is set if needed.
- [ ] Wi-Fi is off before the quiet measurement period.

Five-minute Wi-Fi window

A countdown is shown on the web page. Wi-Fi turns off five minutes after startup, even if a phone remains connected. Select **Keep Wi-Fi On** only when a longer setup session is needed. The choice lasts until the next power cycle.



Closed case



Powered detector



5000 mAh power bank

Wired USB-UART monitor

Use the Mac dashboard while detector Wi-Fi remains off

1. Download and prepare the monitor

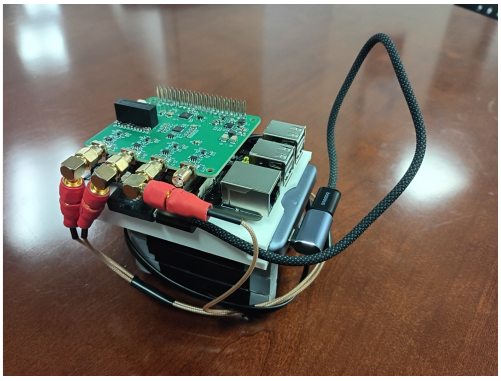
Before the flight, use an internet connection to visit **github.com/tharinduudu/ESP32-P4-gLOWCSOT-cubesat**. Open **dist/Muon-UART-Monitor-macOS.zip** and click **Download raw file**. Unzip it in Downloads. The folder contains **uart_web_monitor.py**.

Python 3 is required on the Mac. In Terminal, run **python3 --version**. If it is unavailable, install Python 3 from **python.org/downloads/macos** before the flight.

2. Connect the detector

Keep the power bank connected to the detector's normal USB-C power input. Connect a data-capable USB cable from the separate **USB-UART** Type-C port to the Mac.

The monitor works with detector Wi-Fi off. Leave the power bank connected for the entire run.



3. Open the monitor from Terminal

1. Open **Terminal** on the Mac. Type **python3** followed by a space; do not press Return yet.
2. In Finder, drag **uart_web_monitor.py** from the unzipped folder into Terminal. The full path will appear.
3. Press Return. Leave Terminal open and visit **http://127.0.0.1:8080** in the Mac's browser.
4. Check **UART connected**, SD status, the detector time, and the latest counts.

Example: `python3 /path/to/uart_web_monitor.py`
This method does not require opening the blocked .command file.

During the run

The browser sends the Mac's time to the detector; confirm the displayed time. The first valid sync renames the active files. Closing Terminal or unplugging the Mac leaves counting and SD logging running on the power bank.

UART unavailable? Check the USB-UART port and data cable. Close screen or other serial programs and restart the monitor. In Terminal, **ls /dev/cu.usbmodem*** lists detected USB serial ports.